A003 v0.3.2

Continuous Topology Safety & Independent Dynamics Qualification

Scope and validated local engineering

104 regression tests: 74 passed, 30 skipped, 0 failed. 13/13 new targeted safety/provenance tests passed. Full new Windows N=1200 execution: NOT RUN in this development environment.

Actual pinned C005 v0.6.1 Python Biot-Savart implementation was exercised on hash-checked original A003 F3/F4 geometries, downsampled from 1200 to N=150. Two independent temporal algorithms were compared at fixed T=2 in source model units.

N=150 local bounded original-atlas checks

Seed	16 step interpolant	RK4 vs DOP853 RMS
T(2,3) F3	PASS (linear only)	2.66e-15
T(3,5) F4	PASS (linear only)	1.72e-14

New gates

G6P14: per-step nonlocal 3-D segment clearance with a sufficient lower bound for linearly interpolated polygon states. G6P15: DOP853 vs RK4 with unchanged authentic C005 velocity operator. G6P16: opt-in provenance-separated KnotPlot/Gilbert computational source families.

The standard 5 quarter-time trajectory NPZ is preserved. An additional 17 states are evaluated per seed and mesh for G6P14; every RK4 substep uses the genuine C005 field on the evolving geometry.

Certification boundary

A certificate for piecewise-linear interpolation between step endpoints is NOT a rigorous bound on the exact continuous ODE solution, and cannot prove continuum Euler topological preservation. G6 stays UNRESOLVED.

Provenance / open gates

G6P8 and G6P9 closure audit | distinct from successful numerical gates

G6P8 - canonical E010 index

A003 creates a candidate source contract and carrier-index JSON, SHA-256-pinned against the original immutable E010 v0.3.1 production contract. The actual E010 native Workbench source resolver accepted all eight frozen A003 analytic carriers in the bounded integration test.

G6P8 is UNRESOLVED because the candidate was deliberately NOT committed into the frozen E010 v0.3.1 source tree. Closure requires a separately versioned E010 canonical registry release, native registered-carrier readback, package/source manifest and freeze checks.

G6P9 - full Euler pressure/velocity/vorticity

The available C005 v0.6.1 Rosenhead-midpoint operator determines an instantaneous regularized filament velocity; its evolving filament RK4 equation is not a 3-D incompressible Euler PDE solver. This release records the absent pressure-Poisson source/boundary solve, momentum residual, physically-calibrated core/vorticity initial conditions, unit calibration and independent continuum solver.

G6P9 remains UNRESOLVED. No mock pressure solve, synthetic PDE residual or full physical qualification is inserted.

Previous real Windows observations (v0.3.1 replay only)

Using quarter-time samples t=0,0.5,1,1.5,2 from the user's v0.3.1 NPZ, the linear-interpolant sufficient certificate passes for N=150/300/600 (both F3/F4). At N=1200 the lower bound is inconclusive. No segment crossing was observed; this does not mean a crossing occurred.

v0.3.2 evaluates every RK4 step endpoint, 16 intervals instead of four. These additional states from a new live Windows execution are not yet available.

Build and execution

Install next to v0.3.1 under 02_libraries/A_knot_libraries/A003_Fourier_Knot_Seed_Atlas. Run `run_all.cmd` for regression/geometry and `run_all.cmd COMPLETE` for full G5->G6. For a source-specific audit, `run_g6_dynamic.cmd --independent-carriers <absolute_manifest_path>`. Exit code 3 can mean scientifically UNRESOLVED even if numerical preconditions PASS.